

TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering and Software Engineering

SENG 491- CMPE 491

Senior Design Project I

PROJECT PROPOSAL

Project Name: EyeNysta: Smartphone-Based Nystagmus Analysis and BPPV Episode Tracking

Project Website: https://zzzeysuuu.github.io/eyenysta/

Team Members:

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Project Description

EyeNysta is an mobile application that we plan to develop for patients who have already been diagnosed with benign paroxysmal positional vertigo (BPPV). Vertigo episodes often occur at home, and the eye movements seen during these episodes may no longer be present by the time the patient visits a doctor. Our aim is to help patients record these episodes using their smartphone cameras so that the recordings can be examined later. The application will guide users through the recording process when it is safe to do so and check whether the video quality is sufficient for analysis. We plan to use image processing and machine learning techniques to track eye movements and measure the direction, slow-phase velocity, and duration of nystagmus. If we can obtain suitable clinical data, we will also investigate whether these measurements can help determine the affected ear canal. The application is not intended to make a medical diagnosis but to provide measurements that doctors can use during their evaluations.

Another part of our project is to keep a record of previous episodes and compare them with new ones. We plan to examine changes in the direction, speed, and duration of nystagmus to identify differences from a patient's usual pattern. This could help doctors follow changes in the patient's condition over time. The application will also allow patients to share reports containing measurement results, graphs, and selected video clips with their doctors. To address safety concerns, we will include a rule-based system that checks for warning symptoms according to clinical guidelines. If any of these symptoms are reported, the application will advise the user to call 112 immediately. However, it will not confirm that an episode is completely safe when no warning signs are reported. We plan to process videos directly on the device and allow users to control when their data are shared. Through this project, we aim to make it easier for patients to document their episodes and for doctors to review the recorded information.

AI Use Statement: AI assistance was used in the preparation of this proposal.